Performance of the FashionMNIST and CIFAR-10 Classifiers

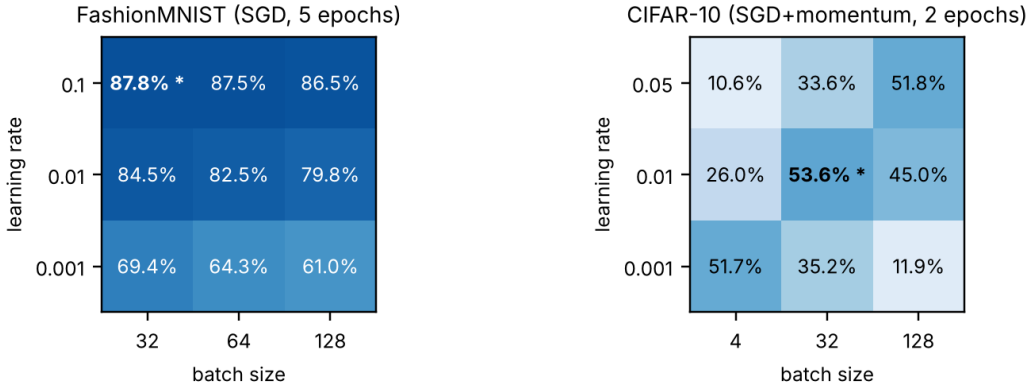
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1. Setup

FashionMNIST (10 garment classes, 28×28 grayscale) uses the fully connected network of the PyTorch Quickstart (784-512-512-10, ~670k parameters). **CIFAR-10** (10 object classes, 32×32 color) uses the small CNN of the PyTorch CIFAR-10 tutorial (2 conv + 3 linear layers, ~62k parameters). Each **baseline** uses the tutorial's settings. For the **tuned** model, the architecture stays identical and only the hyperparameters change: a grid search over learning rate and batch size, with every run starting from the same random weights, scored on a **validation split** of the training data (10k for FashionMNIST, 5k for CIFAR-10). The best combination is then retrained for more epochs on all training images. The test set is used only once, to report the final accuracy, so the choice of hyperparameters never sees it.

2. Results

	Model	Baseline (tutorial)	Tuned	Test accuracy
FashionMNIST	Linear NN	SGD, lr 0.001, batch 64, 5 ep.	SGD, lr 0.1, batch 32, 15 ep.	**64.6% → 89.4% (+24.8)**
CIFAR-10	CNN	SGD+mom., lr 0.001, batch 4, 2 ep.	SGD+mom., lr 0.01, batch 32, 10 ep.	**53.7% → 60.1% (+6.4)**



Validation accuracy of each grid-search run (= chosen). The tutorial settings are lr 0.001 / batch 64 (FashionMNIST) and lr 0.001 / batch 4 (CIFAR-10).*

3. Why hyperparameter optimization matters

- **The same network can be weak or strong.** Without changing a single layer, FashionMNIST accuracy rose from 64.6% to 89.4%. The baseline's training loss was still falling steadily after 5 epochs: the learning rate of 0.001 made the steps too small, so the model was simply under-trained.
- **The learning rate is the most important setting.** On FashionMNIST, moving from 0.001 to 0.1 adds about 20 points for every batch size. Smaller batches also help, because they give more weight updates per epoch.
- **Hyperparameters interact.** On CIFAR-10 the good runs lie on a diagonal: lr 0.001 / batch 4, lr 0.01 / batch 32 and lr 0.05 / batch 128 all reach about 52–54%, because larger batches give less noisy gradients and allow larger steps. Off the diagonal, training fails completely: lr 0.05 with batch 4 diverges (10.6%, chance level), and lr 0.001 with batch 128 barely moves from its random start (11.9%). Tuning one value at a time would miss this.
- **Tuning also improves confidence.** For the same shirt image, the baseline answered "Coat" with 27%, while the tuned model answered "Shirt" with 99%.

4. Limits

- **Tuning cannot exceed the model's capacity.** CIFAR-10 gained only 6.4 points: a 62k-parameter CNN is too small for this harder dataset. Larger gains would need a deeper network (e.g. ResNet), data augmentation or longer training.
- **Similar classes stay confused:** shirt, pullover and coat on FashionMNIST; cat (18.8% for the baseline) and dog on CIFAR-10.
- **Training data shapes behaviour.** A car photographed on a white studio background was classified as "plane", because many CIFAR-10 planes appear against a plain sky. FashionMNIST models also expect light garments on black, so real photos differ strongly from the training data.